

CSCE 5380 DATA MINING

Project 02 Milestone01 Report

TASK 1A: Preprocessing the votes Dataset:

- **Libraries & Dataset Loading:** I started off by loading the arules packages once and then loaded the dataset of Congressional Voting Records and giving all the attributes useful column names from .names file as the .data file did not have any headers. The party affiliation attribute was not excluded in the dataset since it can also be involved in association rule discovery with the attributes of voting.
- **Handling Unknown Values:** In this dataset, the symbol "?" is not the absence of data in the normal meaning. Rather it suggests the cases like abstaining, voting present or not taking a decisive stand. Thus, I did not remove or impute these values. I instead left them as a legitimate type and recategorized them as unknown in order to make it clear.
- **Transforming into Categorical Form:** All the attributes are categorical in the dataset, therefore, I transformed every column into factors. This move is significant in the sense that Apriori algorithm is applied to categorical or transactional data as opposed to numerical data. Factor conversion is the best method of preprocessing as each attribute has a limited number of possible values.
- **Changes into Transaction Format:** Lastly, I transformed the raw processed data to transaction format in the arules package. The reason behind this is that Apriori algorithm assumes data in the form of transactions, where each row contains a number of items. The change rendered the dataset to be applicable in association rule mining.

TASK2B: The Top 20 Rules:

We Extracted the rules using arules on this votes dataset and using the following parameter given and requested by the professor*:

- $\text{supp} = 0.05$ -> low enough to get a reasonable number of rules
- $\text{*conf} = 0.90$ -> as required in the assignment
- $\text{Min length} = 2$ -> at least one item on LHS and one item on RHS
- $\text{Max length} = 17$ -> generates all possible rule combinations.
-

The Apriori model parameters:

*Model: Apriori

Parameter specification:

confidence	support	minlen	maxlen	target	smax	arem	aval	originalSupport	maxtime	ext
0.9	0.05	2	17	rules	1	none	FALSE	TRUE	5	TRUE

Algorithmic control:

filter	tree	heap	memopt	load	sort	verbose
0.1	TRUE	TRUE	FALSE	TRUE	2	TRUE

Absolute minimum support count: 21, The set of **1400172** rules are produced by arules.

Rules Summary:**Length Summary:**

Size	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Count	47	1187	10874	49823	137998	252696	322466	294008	194844	94455	32853	7743	1106	72

Length Statistics:

Metric	Min	1st Qu.	Median	Mean	3rd Qu.	Max
Value	2	7	8	8.297	9	15

Quality Measurements summary:

Metric	Min	1st Qu.	Median	Mean	3rd Qu.	Max
Support	0.05057	0.06207	0.08046	0.09087	0.10115	0.56322
Confidence	0.9	0.9375	0.9655	0.9638	1	1
Coverage	0.05057	0.06437	0.08276	0.09453	0.10575	0.61379
Lift	1.439	1.711	1.926	1.967	2.256	2.862
Count	22	27	35	39.53	44	245

The Top 20 Rules Extracted:

Sl. No.	Rule No. in Rules Df	rules	support	confidence	coverage	lift	count
1	75165	{handicapped_infants=y,el_salvador_aid=n,immigration=n,crime=n,export_administration_act_south_africa=unknown}=>{religious_groups_in_schools=n}	0.05977	1	0.05977	2.861842	26
2	75290	{el_salvador_aid=n,immigration=n,supervend_right_to_sue=n,crime=n,export_administration_act_south_africa=unknown}=>{religious_groups_in_schools=n}	0.064368	1	0.064368	2.861842	28
3	75545	{physician_fee_freeze=n,el_salvador_aid=n,immigration=n,crime=n,export_administration_act_south_africa=unknown}=>{religious_groups_in_schools=n}	0.068966	1	0.068966	2.861842	30
4	75549	{adoption_of_budget_resolution=y,el_salvador_aid=n,immigration=n,crime=n,export_administration_act_south_africa=	0.066667	1	0.066667	2.861842	29

		unknown} {religious_groups_in_schools=n} =>					
5	223024	{el_salvador_aid=n,immigration=n,su rfund_right_to_sue=n,crime=n,duty_fre e_exports=y,export_administration_act_ south_africa=unknown} => {religious_groups_in_schools=n}	0.050575	1	0.050575	2.861842	22
6	223208	{physician_fee_freeze=n,el_salvador_ai d=n,immigration=n,crime=n,duty_free_ exports=y,export_administration_act_so uth_africa=unknown} => {religious_groups_in_schools=n}	0.052874	1	0.052874	2.861842	23
7	223212	{adoption_of_budget_resolution=y,el_sa lvador_aid=n,immigration=n,crime=n,d uty_free_exports=y,export_administrati on_act_south_africa=unknown} => {religious_groups_in_schools=n}	0.050575	1	0.050575	2.861842	22
8	223449	{handicapped_infants=y,el_salvador_ai d=n,immigration=n,superfund_right_to_ sue=n,crime=n,export_administration_ act_south_africa=unknown} => {religious_groups_in_schools=n}	0.057471	1	0.057471	2.861842	25
9	223566	{handicapped_infants=y,el_salvador_ai d=n,mx_missile=y,immigration=n,crime =n,export_administration_act_south_afr ica=unknown} => {religious_groups_in_schools=n}	0.052874	1	0.052874	2.861842	23
10	223656	{handicapped_infants=y,el_salvador_ai d=n,immigration=n,education_spendin g=n,crime=n,export_administration_act_ south_africa=unknown} => {religious_groups_in_schools=n}	0.05977	1	0.05977	2.861842	26
11	223660	{handicapped_infants=y,el_salvador_ai d=n,anti_satellite_test_ban=y,immigrati on=n,crime=n,export_administration_a ct_south_africa=unknown} => {religious_groups_in_schools=n}	0.05977	1	0.05977	2.861842	26
12	223664	{handicapped_infants=y,el_salvador_ai d=n,aid_to_nicaraguan_contras=y,immi gration=n,crime=n,export_administratio n_act_south_africa=unknown} => {religious_groups_in_schools=n}	0.05977	1	0.05977	2.861842	26
13	223668	{handicapped_infants=y,physician_fee_ freeze=n,el_salvador_aid=n,immigratio n=n,crime=n,export_administration_act_ south_africa=unknown} => {religious_groups_in_schools=n}	0.05977	1	0.05977	2.861842	26
14	223672	{handicapped_infants=y,adoption_of_b udget_resolution=y,el_salvador_aid=n,i mmigration=n,crime=n,export_administ	0.057471	1	0.057471	2.861842	25

		ration_act_south_africa=unknown} => {religious_groups_in_schools=n}					
15	223676	{party=democrat,handicapped_infants=y,el_salvador_aid=n,immigration=n,crime=n,export_administration_act_south_africa=unknown} => {religious_groups_in_schools=n}	0.05977	1	0.05977	2.861842	26
16	223903	{el_salvador_aid=n,mx_missile=y,immigration=n,superfund_right_to_sue=n,crime=n,export_administration_act_south_africa=unknown} => {religious_groups_in_schools=n}	0.055172	1	0.055172	2.861842	24
17	224037	{el_salvador_aid=n,immigration=n,education_spending=n,superfund_right_to_sue=n,crime=n,export_administration_act_south_africa=unknown} => {religious_groups_in_schools=n}	0.064368	1	0.064368	2.861842	28
18	224042	{el_salvador_aid=n,anti_satellite_test_ban=y,immigration=n,superfund_right_to_sue=n,crime=n,export_administration_act_south_africa=unknown} => {religious_groups_in_schools=n}	0.064368	1	0.064368	2.861842	28
19	224047	{el_salvador_aid=n,aid_to_nicaraguan_contras=y,immigration=n,superfund_right_to_sue=n,crime=n,export_administration_act_south_africa=unknown} => {religious_groups_in_schools=n}	0.064368	1	0.064368	2.861842	28
20	224052	{physician_fee_freeze=n,el_salvador_aid=n,immigration=n,superfund_right_to_sue=n,crime=n,export_administration_act_south_africa=unknown} => {religious_groups_in_schools=n}	0.064368	1	0.064368	2.861842	28

Task 3B: Redundant Rules:

I have identified the redundant rules in the top 20 rules that we extracted and removed them. With association rule mining, a rule is said to be redundant when there is an already existing general rule with the same consequent and with an equal or greater confidence. In such a case, the more specific rule does not bring something new. Most of the rules shared the same consequent in my top 20 rules i.e., Religious_groups_in_schools = n. Some of these rules were also sharing the same confidence value of 1.0 as well as sharing the same lift value of approximately 2.8618. They were simply the same except that additional items were put on the left-hand. As the extra conditions did not enhance the confidence, such rules were unnecessary.

Redundant Rules:

The following rules were removed as redundant because they were more specific versions of stronger or equally strong general rules i.e., subset rules with same or higher confidence:

Rule 223024: {el_salvador_aid=n, immigration=n, superfund_right_to_sue=n, crime=n, duty_free_exports=y, export_administration_act_south_africa=unknown} => {religious_groups_in_schools=n} has same confidence as rule 75290 and rule 75290 is subset of 223024

Similarly the rules below:

Rule 223208: {physician_fee_freeze=n, el_salvador_aid=n, immigration=n, crime=n, duty_free_exports=y, export_administration_act_south_africa=unknown} => {religious_groups_in_schools=n}

Rule 223212: {adoption_of_budget_resolution=y, el_salvador_aid=n, immigration=n, crime=n, duty_free_exports=y, export_administration_act_south_africa=unknown} => {religious_groups_in_schools=n}

Rule 223449: {handicapped_infants=y, el_salvador_aid=n, immigration=n, superfund_right_to_sue=n, crime=n, export_administration_act_south_africa=unknown} => {religious_groups_in_schools=n}

- These rules ranking 5,6,7,8 have a general rule or subset rule in the ranking of 2,3,4,1 respectively. Likewise remaining 9-20 ranked rules also have the subset rule with same confidence so these rules(5-20) will be redundant and the **respective rule that is reason for this redundancy** is noted in the table below:

Top20_Position	Rule_Id	Rule	Is_Redundant	Source_Rule_Position	Source_Rule_Id	Source_Rule	Support	Confidence	Coverage	Lift	Count
1	75165	{handicapped_infants=y,el_salvador_aid=n,immigration=n,crime=n,export_administration_act_south_africa=unknown}=> {religious_groups_in_schools=n}	FALSE	NA	NA	NA	0.05977	1	0.05977	2.861842	26
2	75290	{el_salvador_aid=n,immigration=n,superfund_right_to_sue=n,crime=n,export_administration_act_south_africa=unknown}=> {religious_groups_in_schools=n}	FALSE	NA	NA	NA	0.064368	1	0.064368	2.861842	28

3	7 5 5 4 5	{physician_fee_freeze=n,el_salvador_aid=n,immigration=n,crime=n,export_administration_act_south_africa=unknown}=> {religious_groups_in_schools=n}	F A L S E	N A	N A	NA	0 . 0 6 8 9 6 6	1	0. 0 6 8 9 6 6	2. 8 6 1 8 4 2	3 0
4	7 5 5 4 9	{adoption_of_budget_resolution=y,el_salvador_aid=n,immigration=n,crime=n,export_administration_act_south_africa=unknown}=> {religious_groups_in_schools=n}	F A L S E	N A	N A	NA	0 . 0 6 6 6 6 6 7	1	0. 0 6 6 6 6 7	2. 8 6 1 8 4 2	2 9
5	2 2 3 0 2 4	{el_salvador_aid=n,immigration=n,superfund_right_to_sue=n,crime=n,duty_free_exports=y,export_administration_act_south_africa=unknown}=> {religious_groups_in_schools=n}	T R U E	2	7 5 2 9 0	{el_salvador_aid=n,immigration=n,superfund_right_to_sue=n,crime=n,export_administration_act_south_africa=unknown}=> {religious_groups_in_schools=n}	0 . 0 5 0 5 7 5	1	0. 0 5 0 5 7 5	2. 8 6 1 8 4 2	2 2
6	2 2 3 2 0 8	{physician_fee_freeze=n,el_salvador_aid=n,immigration=n,crime=n,duty_free_exports=y,export_administration_act_south_africa=unknown}=> {religious_groups_in_schools=n}	T R U E	3	7 5 5 4 5	{physician_fee_freeze=n,el_salvador_aid=n,immigration=n,crime=n,export_administration_act_south_africa=unknown}=> {religious_groups_in_schools=n}	0 . 0 5 2 8 7 4	1	0. 0 5 2 8 7 4	2. 8 6 1 8 4 2	2 3
7	2 2 3 2 1 2	{adoption_of_budget_resolution=y,el_salvador_aid=n,immigration=n,crime=n,duty_free_exports=y,export_administration_act_south_africa=unknown}=> {religious_groups_in_schools=n}	T R U E	4	7 5 5 4 9	{adoption_of_budget_resolution=y,el_salvador_aid=n,immigration=n,crime=n,export_administration_act_south_africa=unknown}=> {religious_groups_in_schools=n}	0 . 0 5 0 5 7 5	1	0. 0 5 0 5 7 5	2. 8 6 1 8 4 2	2 2
8	2 2 3 4 4 9	{handicapped_infants=y,el_salvador_aid=n,immigration=n,superfund_right_to_sue=n,crime=n,export_administration_act_south_africa=unknown}=> {religious_groups_in_schools=n}	T R U E	1	7 5 1 6 5	{handicapped_infants=y,el_salvador_aid=n,immigration=n,crime=n,export_administration_act_south_africa=unknown}=> {religious_groups_in_schools=n}	0 . 0 5 7 4 7 1	1	0. 0 5 7 4 7 1	2. 8 6 1 8 4 2	2 5
9	2 2	{handicapped_infants=y,el_salvador_aid=n,mx_missile=y,immigration=n,crime=n,export_administration_act_south_africa=unknown}=> {religious_groups_in_schools=n}	T R	1	7 5	{handicapped_infants=y,el_salvador_aid=n,immigration=n,crime=n,export_administration_act_south_africa=unknown}=> {religious_groups_in_schools=n}	0 .	1	0. 0	2. 8	2 3

	3 5 6 6	rime=n,export_administration_act_south_africa=unknown}=> {religious_groups_in_schools=n}	U E		1 6 5	me=n,export_administration_act_south_africa=unknown}=> {religious_groups_in_schools=n}	0 5 2 8 7 4		5 2 8 7 4	6 1 8 4 2	
1 0	2 2 3 6 5 6	{handicapped_infants=y,el_salvador_aid=n,immigration=n,education_speending=n,crime=n,export_administration_act_south_africa=unknown}=> {religious_groups_in_schools=n}	T R U E	1	7 5 1 6 5	{handicapped_infants=y,el_salvador_aid=n,immigration=n,crime=n,export_administration_act_south_africa=unknown}=> {religious_groups_in_schools=n}	0 . 0 5 9 7 7	1	0. 0 5 9 7 7	2. 8 6 1 8 4 2	2 6
1 1	2 2 3 6 6 0	{handicapped_infants=y,el_salvador_aid=n,anti_satellite_test_ban=y,immigration=n,crime=n,export_administration_act_south_africa=unknown}=> {religious_groups_in_schools=n}	T R U E	1	7 5 1 6 5	{handicapped_infants=y,el_salvador_aid=n,immigration=n,crime=n,export_administration_act_south_africa=unknown}=> {religious_groups_in_schools=n}	0 . 0 5 9 7 7	1	0. 0 5 9 7 7	2. 8 6 1 8 4 2	2 6
1 2	2 2 3 6 6 4	{handicapped_infants=y,el_salvador_aid=n,aid_to_nicaraguan_contras=y,immigration=n,crime=n,export_administration_act_south_africa=unknown}=> {religious_groups_in_schools=n}	T R U E	1	7 5 1 6 5	{handicapped_infants=y,el_salvador_aid=n,immigration=n,crime=n,export_administration_act_south_africa=unknown}=> {religious_groups_in_schools=n}	0 . 0 5 9 7 7	1	0. 0 5 9 7 7	2. 8 6 1 8 4 2	2 6
1 3	2 2 3 6 6 8	{handicapped_infants=y,physician_freeze=n,el_salvador_aid=n,immigration=n,crime=n,export_administration_act_south_africa=unknown}=> {religious_groups_in_schools=n}	T R U E	1	7 5 1 6 5	{handicapped_infants=y,el_salvador_aid=n,immigration=n,crime=n,export_administration_act_south_africa=unknown}=> {religious_groups_in_schools=n}	0 . 0 5 9 7 7	1	0. 0 5 9 7 7	2. 8 6 1 8 4 2	2 6
1 4	2 2 3 6 7 2	{handicapped_infants=y,adoption_of_budget_resolution=y,el_salvador_aid=n,immigration=n,crime=n,export_administration_act_south_africa=unknown}=> {religious_groups_in_schools=n}	T R U E	1	7 5 1 6 5	{handicapped_infants=y,el_salvador_aid=n,immigration=n,crime=n,export_administration_act_south_africa=unknown}=> {religious_groups_in_schools=n}	0 . 0 5 7 4 7 1	1	0. 0 5 7 4 7 1	2. 8 6 1 8 4 2	2 5
1 5	2 2 3 6 7 6	{party=democrat,handicapped_infants=y,el_salvador_aid=n,immigration=n,crime=n,export_administration_act_south_africa=unknown}=> {religious_groups_in_schools=n}	T R U E	1	7 5 1 6 5	{handicapped_infants=y,el_salvador_aid=n,immigration=n,crime=n,export_administration_act_south_africa=unknown}=> {religious_groups_in_schools=n}	0 . 0 5 9 7 7	1	0. 0 5 9 7 7	2. 8 6 1 8 4 2	2 6

16	223903	{el_salvador_aid=n,mx_missile=y,immigration=n,superfund_right_to_sue=n,crime=n,export_administration_act_south_africa=unknown}=>{religious_groups_in_schools=n}	TRUE	2	75290	{el_salvador_aid=n,immigration=n,superfund_right_to_sue=n,crime=n,export_administration_act_south_africa=unknown}=>{religious_groups_in_schools=n}	0.055172	1	0.5172	2.8142	24
17	224037	{el_salvador_aid=n,immigration=n,education_spending=n,superfund_right_to_sue=n,crime=n,export_administration_act_south_africa=unknown}=>{religious_groups_in_schools=n}	TRUE	2	75290	{el_salvador_aid=n,immigration=n,superfund_right_to_sue=n,crime=n,export_administration_act_south_africa=unknown}=>{religious_groups_in_schools=n}	0.064368	1	0.64368	2.8142	28
18	224042	{el_salvador_aid=n,anti_satellite_test_ban=y,immigration=n,superfund_right_to_sue=n,crime=n,export_administration_act_south_africa=unknown}=>{religious_groups_in_schools=n}	TRUE	2	75290	{el_salvador_aid=n,immigration=n,superfund_right_to_sue=n,crime=n,export_administration_act_south_africa=unknown}=>{religious_groups_in_schools=n}	0.064368	1	0.64368	2.8142	28
19	224047	{el_salvador_aid=n,aid_to_nicaragua_n_contras=y,immigration=n,superfund_right_to_sue=n,crime=n,export_administration_act_south_africa=unknown}=>{religious_groups_in_schools=n}	TRUE	2	75290	{el_salvador_aid=n,immigration=n,superfund_right_to_sue=n,crime=n,export_administration_act_south_africa=unknown}=>{religious_groups_in_schools=n}	0.064368	1	0.64368	2.8142	28
20	224052	{physician_fee_freeze=n,el_salvador_aid=n,immigration=n,superfund_right_to_sue=n,crime=n,export_administration_act_south_africa=unknown}=>{religious_groups_in_schools=n}	TRUE	2	75290	{el_salvador_aid=n,immigration=n,superfund_right_to_sue=n,crime=n,export_administration_act_south_africa=unknown}=>{religious_groups_in_schools=n}	0.064368	1	0.64368	2.8142	28

As We can see from the table the reason why the each of the rules ranking 5-20 are redundant as they same confidence i.e., 1 as the subset rules ranking 1-4. The table also contains the rule ids of the rules dataframe that we generated previously in R code file. Now We can say that reasons are provided for the redundant rules in the TOP 20. Getting all the redundant rules with source rule for all 1.3million rules is not efficient and code will be $O(n^2)$ Time complexity, we are focusing only on the set like top 20. So We are not working on the complete rules.

Then Now lets see the **Top20 Non Redundant** rules:

Rank	Rule_Id	Rule	Support	Confidence	Coverage	Lift	Count
1	75165	{handicapped_infants=y,el_salvador_aid=n,immigration=n,crime=n,export_administration_act_south_africa=unknown} => {religious_groups_in_schools=n}	0.05977	1	0.05977	2.861842	26
2	75290	{el_salvador_aid=n,immigration=n,superfund_right_to_sue=n,crime=n,export_administration_act_south_africa=unknown} => {religious_groups_in_schools=n}	0.064368	1	0.064368	2.861842	28
3	75545	{physician_fee_freeze=n,el_salvador_aid=n,immigration=n,crime=n,export_administration_act_south_africa=unknown} => {religious_groups_in_schools=n}	0.068966	1	0.068966	2.861842	30
4	75549	{adoption_of_budget_resolution=y,el_salvador_aid=n,immigration=n,crime=n,export_administration_act_south_africa=unknown} => {religious_groups_in_schools=n}	0.066667	1	0.066667	2.861842	29
34	225472	{el_salvador_aid=n,immigration=n,education_spending=n,superfund_right_to_sue=n,duty_free_exports=y,export_administration_act_south_africa=unknown} => {religious_groups_in_schools=n}	0.050575	1	0.050575	2.861842	22
35	225787	{physician_fee_freeze=n,el_salvador_aid=n,immigration=n,education_spending=n,duty_free_exports=y,export_administration_act_south_africa=unknown} => {religious_groups_in_schools=n}	0.052874	1	0.052874	2.861842	23
36	225791	{adoption_of_budget_resolution=y,el_salvador_aid=n,immigration=n,education_spending=n,duty_free_exports=y,export_administration_act_south_africa=unknown} => {religious_groups_in_schools=n}	0.050575	1	0.050575	2.861842	22
601	75533	{el_salvador_aid=n,immigration=n,education_spending=n,crime=n,export_administration_act_south_africa=unknown} => {religious_groups_in_schools=n}	0.068966	0.967742	0.071264	2.769525	30
602	75537	{el_salvador_aid=n,anti_satellite_test_ban=y,immigration=n,crime=n,export_administration_act_south_africa=unknown} => {religious_groups_in_schools=n}	0.068966	0.967742	0.071264	2.769525	30

603	75553	{party=democrat,el_salvador_aid=n,immigration=n,crime=n,export_administration_act_south_africa=unknown} => {religious_groups_in_schools=n}	0.068966	0.967742	0.071264	2.769525	30
604	75658	{physician_fee_freeze=n,anti_satellite_test_ban=y,immigration=n,crime=n,export_administration_act_south_africa=unknown} => {religious_groups_in_schools=n}	0.068966	0.967742	0.071264	2.769525	30
614	227734	{physician_fee_freeze=n,el_salvador_aid=n,anti_satellite_test_ban=y,immigration=n,education_spending=n,export_administration_act_south_africa=unknown} => {religious_groups_in_schools=n}	0.068966	0.967742	0.071264	2.769525	30
627	75662	{adoption_of_budget_resolution=y,anti_satellite_test_ban=y,immigration=n,crime=n,export_administration_act_south_africa=unknown} => {religious_groups_in_schools=n}	0.066667	0.966667	0.068966	2.766447	29
632	227739	{adoption_of_budget_resolution=y,el_salvador_aid=n,anti_satellite_test_ban=y,immigration=n,education_spending=n,export_administration_act_south_africa=unknown} => {religious_groups_in_schools=n}	0.066667	0.966667	0.068966	2.766447	29
651	75331	{anti_satellite_test_ban=y,immigration=n,superfund_right_to_sue=n,crime=n,export_administration_act_south_africa=unknown} => {religious_groups_in_schools=n}	0.064368	0.965517	0.066667	2.763158	28
652	76559	{el_salvador_aid=n,immigration=n,education_spending=n,superfund_right_to_sue=n,export_administration_act_south_africa=unknown} => {religious_groups_in_schools=n}	0.064368	0.965517	0.066667	2.763158	28
661	251727	{handicapped_infants=y,water_project_cost_sharing=n,el_salvador_aid=n,synfuels_corporation_cutback=n,superfund_right_to_sue=n,crime=n} => {religious_groups_in_schools=n}	0.064368	0.965517	0.066667	2.763158	28
662	251790	{handicapped_infants=y,water_project_cost_sharing=n,aid_to_nicaraguan_contras=y,synfuels_corporation_cutback=n,superfund_right_to_sue=n,crime=n} => {religious_groups_in_schools=n}	0.064368	0.965517	0.066667	2.763158	28
697	251698	{handicapped_infants=y,water_project_cost_sharing=n,mx_missile=y,synfuels_corporation_cutback=n,superfund_right_to_sue=n,crime=n} => {religious_groups_in_schools=n}	0.062069	0.964286	0.064368	2.759633	27

		ght_to_sue=n,crime=n} {religious_groups_in_schools=n}=>					
755	17177	{handicapped_infants=y,immigration= n,crime=n,export_administration_act_ south_africa=unknown} {religious_groups_in_schools=n}=>	0.05977	0.962963	0.062069	2.755848	26